LeBert Sam Billgates, PhD

National Institute of Technology, Tiruchirappalli, India

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Date of Birth: 6th July 1995 **| Age:** 30 **| Nationality:** Indian

Highlights & Interest

- Experimental and Computational Physics, Magnetic materials, Exchange bias, Low-temperature magnetic studies, Positron Annihilation Spectroscopy, Monte Carlo Spin Dynamics, ML Aided Material Prediction, Agentic AI tools for research, autonomous experimental and computation workflow.

Qualification

SRM University, Andhra Pradesh, India **2025-2026 (Aug)**

Post Doctoral Fellow, Center for Interdisciplinary Research

Project: Accelerating data driven research by automating structured data extraction using LLMs.

National Institute of Technology (NIT), Tiruchirappalli, India **2019-2024 (Dec)**

Ph.D. in Physics

Thesis title: '*Structural and finite temperature magnetism of non-equilibrium FeNi Alloys'*

Loyola College (University of Madras), Chennai, India **2016-2018**

Master's in Physics

Loyola College (University of Madras), Chennai, India **2013-2016**

Bachelor's in Physics

Research Experience

Center for Interdisciplinary Research, SRM University AP **2025-2026**

Supervisor: Prof. Mahesh Kumar Ravva

My postdoctoral research work is on developing a multi-agent AI tool for automated structured data extraction from scientific articles. I have got good experience in architecting and building agentic AI system from scratch. This could help me in building autonomous laboratory setups.

Magnetic materials laboratory, NIT Tiruchirappalli **2019-2024**

Supervisor: Prof. R. Justin Joseyphus

My research focused on understanding the non-equilibrium bcc phase stabilization and finite temperature magnetic behavior of FeNi alloys synthesized through polyol mediated reaction. My work combines experimental studies and computational modeling with machine learning integration,

- The influence of cluster vacancies on the bcc phase stabilization in FeNi alloys is studied using Positron Annihilation Spectroscopy.
- A new computational framework is developed to predict the near-neighboring exchange interactions by coupling spin dynamics simulation and machine learning models.
- The unusual temperature-dependent magnetization (anisotropy) behavior of polyol synthesized FeNi alloys are modelled using Monte Carlo (Constrained Monte Carlo) spin dynamics by including the effect of cluster vacancies.

- The Curie temperature of FeNi Invar alloy is enhanced by 65% (largest ever synthesized) compared to the bulk alloy and the origin of Invar behavior is studied using spin dynamics by considering longitudinal spin fluctuations.

Experimental expertise

- Solid handling experience of Transmission Electron Microscopy (TEM)
- Temperature dependent magnetic measurements using Vibrational Sample Magnetometer (VSM)
- Converted a low-cost unipolar power supply to bipolar using integrated circuits when the bipolar power supply of VSM failed
- Thermal analysis facilities such as TGA/DTA

Computational expertise

- Atomistic spin dynamics (Monte Carlo, Constrained Monte Carlo)
- Machine learning aided material prediction
- Basic *ab initio* calculations

Intern at Kodaikanal Solar observatory, Indian Institute of Astrophysics **2017**

Basic telescope operations and observation techniques.

Publications

- Vismaya Rajeevan, **LeBert Sam Billgates** and Raphel Justin Joseyphus Interfacial anisotropy induced enhanced exchange bias in FeCo Nanoparticles, *Physica Status Solidi A,* 2025.
- **LeBert Sam Billgates,** Vismaya Rajeevan and Raphel Justin Joseyphus Temperature-dependent magnetic properties of bcc and fcc FeNi alloys: A machine learning assisted Monte-Carlo approach, *Physica B Condensed Matter,* 692, 2024
- **LeBert Sam Billgates** and Raphel Justin Joseyphus Influence of vacancy type defects on the exchange integrals in ferromagnetic metals: a Monte-Carlo-based approach, *Physica Scripta*, 2024
- **LeBert Sam Billgates,** George Antilen Jacob, Selvakumar Sellaiyan and Raphel Justin Joseyphus Unconventional Nonequilibrium Phase Stabilization in FeNi Alloy Examined by Positron Annihilation Spectroscopy, *Physica Status Solidi B*, 2023
- Vismaya Rajeevan, Selvakumar Selliyan, **LeBert Sam Billgates**, R Justin Joseyphus, Disordered phase and vacancies in FeCo alloy analyzed from Doppler broadening and Mössbauer studies, *Physica Scripta*, 2025
- **LeBert Sam Billgates** and Raphel Justin Joseyphus, Transverse Spin Fluctuations in Polyol-Synthesized FeNi Invar Alloy with Large Curie Temperature *(under submission)*
- **LeBert Sam Billgates,** Mahesh Kumar Ravva, Anita Halder, Piper: A Multi-Agent Framework For Automated Chemical Data Extraction From Research Articles *(under submission)*

Technical Skills & abilities

- **Programming:** Python, Basic C++, Bash
- **Data Analysis:** Matplotlib, Power BI, SciKit
- **Other Tools:** Git, LabVIEW, Instrumentation, Adobe Illustrator/Photoshop, HPC
- **Language:** Fluent in English and Tamil

Projects

Name of the Startup: SciX (MSME)

Description: Reference managing tool (Incorporates web scraping tools) and Event management serivces

Grants received: 150000 INR (seed grant under ‘SIEMENS Startup Innovation’)

Incubation Centre: Centre for Entrepreneurship Development and Incubation (CEDI NITT)

Awards & Honors

- President of SPIE student chapter NITT during 2023-2024
- Student advisor for Nakshatra (Astronomy) club of NITT and contributed to the development of 21cm radio telescope

Certifications

1. Data Analytics with Power BI (2024) - *Codebasics*
2. Synergistic Training Program Utilizing the Scientific and Technological Infrastructure (2023) **(For handling one-week hands-on training program on TEM and VSM)** - *Department of Science and Technology (DST), Govt. of India*
3. Fundamentals of deep learning (2021) - *NVIDIA DLI*
4. Fundamentals of Accelerated Data Science with RAPIDS (2021) - *NVIDIA DLI*
5. 100 hours certificate course in Astronomy and Astrophysics (2017) - *M.P. Birla Institute of Fundamental research, Bangalore*

Conferences & Workshops

- Tamil Nadu State Conclave (Startup Initiative) (2022) - *Software Technology Parks of India*
- Workshop on High Performance Computing (HPC) (2022) - *CDAC & National Institute of Technology, Tiruchirappalli*
- School on Microscopic Characterization Techniques (TEM/SEM/AFM) (2021) - *Inter-University Accelerator Centre, New Delhi, India*
- Around-the-Clock Around-the Globe (AtC-AtG) Magnetics Conference (2021)
- Basic Training Program in Nano Science and Technology (2019) - *Indian Institute of Science and Technology, Bangalore, India*

Reference

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